

systemd

system initialization process

- the default for Red Hat and Debian distros
- instead of lots of small initialization scripts, it uses one program that uses configuration files for each service

Units and Targets

unit and targets

units

- what you operate on and manage
 - defines a service
 - consists of name, type, configuration file
- e.g.*
- automount
 - device
 - mount
 - path
 - service
 - snapshot
 - socket
 - target

targets

group units together

systemctl

system control

```
systemctl [options] command [unit]
```

example

```
# systemctl list-units
```

UNIT	LOAD	ACTIVE	SUB	DESCRIPTION
...				
crond.service	loaded	active	running	Command Scheduler
cups.service	loaded	active	running	CUPS Printing Service
dbus.service	loaded	active	running	D-Bus System Message
...				
multi-user.target	loaded	active	active	Multi-User System
network.target	loaded	active	active	Network

e.g. network.target groups all the units needed to start the network interfaces for the system

- instead of changing runlevels to alter what's running, you change **targets**
- called **runlevel0.target** - **runlevel6.target**

Configuring units

Configuration file

each unit requires a configuration file

defines what program it starts

how it should be started

files are stored in `/lib/systemd/system`

Service

```
# cat sshd.service
[Unit]
Description=OpenSSH server daemon
After=syslog.target network.target auditd.service

[Service]
EnvironmentFile=/etc/sysconfig/ssh
ExecStartPre=/usr/sbin/ssh-keygen
ExecStart=/usr/sbin/ssh -D $OPTIONS
ExecReload=/bin/kill -HUP $MAINPID
KillMode=process
Restart=on-failure
RestartSec=42s

[Install]
WantedBy=multi-user.target
#
```

/usr/sbin/sshd program to run

After=... services to run after

WantedBy=... target level the system should be in

Restart=... how to reload the program

Target

define which service units to start

```
# cat graphical.target
# This file is part of systemd.
#
# systemd is free software; you can redistribute it and/or modify it
# under the terms of the GNU Lesser General Public License as published by
# the Free Software Foundation; either version 2.1 of the License, or
# (at your option) any later version.
[Unit]
Description=Graphical Interface
Documentation=man:systemd.special(7)
Requires=multi-user.target
After=multi-user.target
Conflicts=rescue.target
Wants=display-manager.service
AllowIsolate=yes

[Install]
Alias=default.target
#
```

Requires, After, Conflicts

Setting Default Target

The default target when Linux boots is `/etc/systemd/system/default.target`

it's normally a link to a file in `/lib/systemd/system`

systemctl

system control

control services and targets

`systemctl [options] command [unit]`**commands**

<code>list-units</code>	
<code>start name</code>	
<code>stop name</code>	
<code>reload name</code>	reload unit configuration file
<code>restart name</code>	shut down and restart
<code>status name</code>	
<code>enable name</code>	
<code>disable name</code>	

example

```
james@laptop:/lib/systemd/system$ systemctl status nvidia-persistenced.service
● nvidia-persistenced.service - NVIDIA Persistence Daemon
   Loaded: loaded (/usr/lib/systemd/system/nvidia-persistenced.service; static)
   Active: active (running) since Mon 2025-07-28 08:16:56 NZST; 5h 27min ago
 Main PID: 1628 (nvidia-persiste)
    Tasks: 1 (limit: 18722)
   Memory: 980.0K (peak: 1.4M)
      CPU: 3ms
   CGroup: /system.slice/nvidia-persistenced.service
           └─1628 /usr/bin/nvidia-persistenced --user nvidia-persistenced --no-persistence-mode --verbose

Jul 28 08:16:56 laptop systemd[1]: Starting nvidia-persistenced.service - NVIDIA Persistence Daemon...
Jul 28 08:16:56 laptop nvidia-persistenced[1628]: Verbose syslog connection opened
Jul 28 08:16:56 laptop nvidia-persistenced[1628]: Now running with user ID 122 and group ID 124
Jul 28 08:16:56 laptop nvidia-persistenced[1628]: Started (1628)
Jul 28 08:16:56 laptop nvidia-persistenced[1628]: device 0000:01:00.0 - registered
Jul 28 08:16:56 laptop nvidia-persistenced[1628]: Local RPC services initialized
Jul 28 08:16:56 laptop systemd[1]: Started nvidia-persistenced.service - NVIDIA Persistence Daemon.
```